KAIZEN MATHS

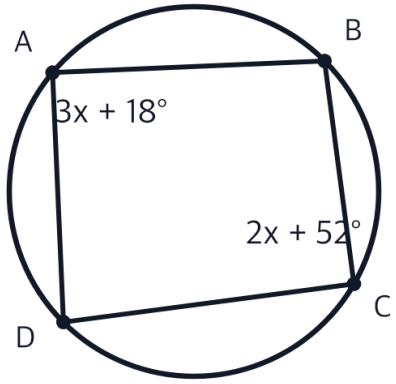
Lesson starter

Circle Theorems

Algebra & Proof

Name _____

SA1



ABCD is cyclic. Opposite angles are 3x + 18° and 2x + 52°. Find x.

GEO-475F-L3 · SA

Show your working

KAIZEN MATHS

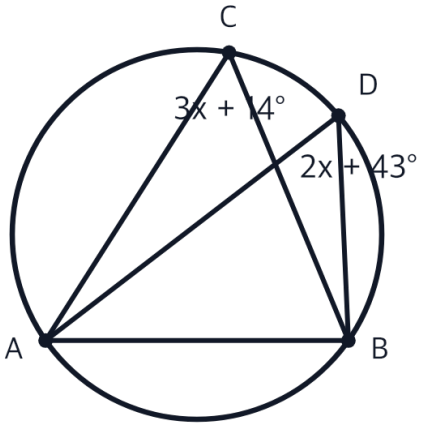
Lesson starter

Circle Theorems

Algebra & Proof

Name _____

SB1



Angles ∠ACB and ∠ADB stand on chord AB. If ∠ACB = 3x + 14° and ∠ADB = 2x + 43°, find x.

GEO-475F-L3 · SB

Show your working

KAIZEN MATHS

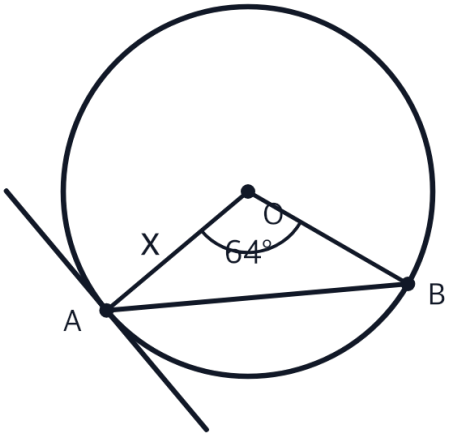
Lesson starter

Circle Theorems

Algebra & Proof

Name _____

SC1



OA and OB are radii. The line at A is a tangent and ∠AOB = 64°. Find the angle between the tangent and chord AB.

GEO-475F-L3 · SC

Show your working

KAIZEN MATHS

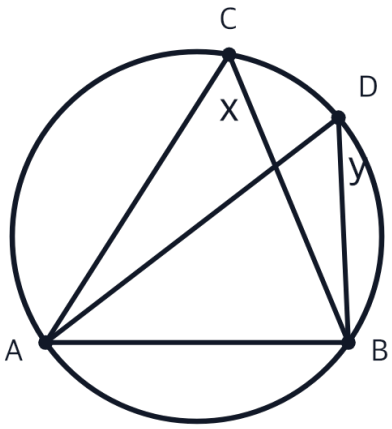
Lesson starter

Circle Theorems

Algebra & Proof

Name _____

SD1



Explain why ∠ACB = ∠ADB.

GEO-475F-L3 · SD

Show your working

KAIZEN MATHS

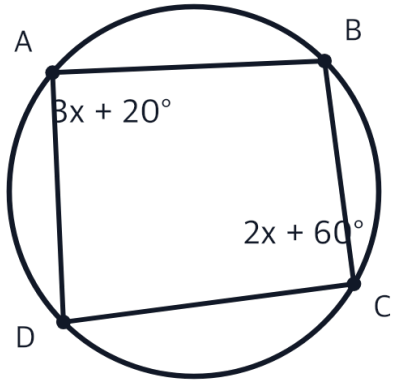
Exit ticket

Circle Theorems

Algebra & Proof

Name _____

EA1



ABCD is cyclic. Opposite angles are 3x + 20° and 2x + 60°. Find x.

GEO-475F-L3 · EA

Show your working

KAIZEN MATHS

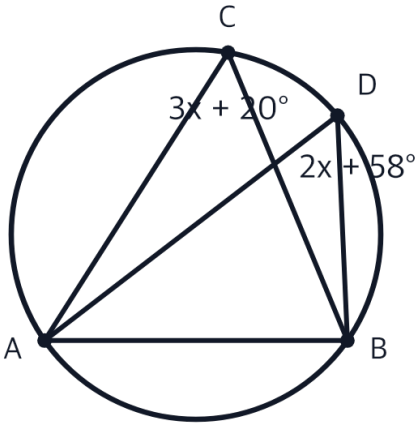
Exit ticket

Circle Theorems

Algebra & Proof

Name _____

EB1



Angles ∠ACB and ∠ADB stand on chord AB. If ∠ACB = 3x + 20° and ∠ADB = 2x + 58°, find x.

GEO-475F-L3 · EB

Show your working

KAIZEN MATHS

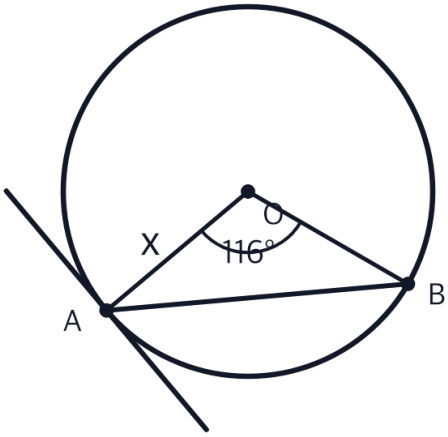
Exit ticket

Circle Theorems

Algebra & Proof

Name _____

EC1



OA and OB are radii. The line at A is a tangent and ∠AOB = 116°. Find the angle between the tangent and chord AB.

GEO-475F-L3 · EC

Show your working

KAIZEN MATHS

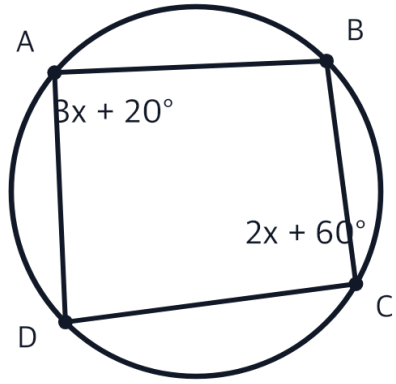
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Circle Theorems

Algebra & Proof

Name _____

EA1



ABCD is cyclic. Opposite angles are 3x + 20° and 2x + 60°. Find x.

GEO-475F-L3 · EA

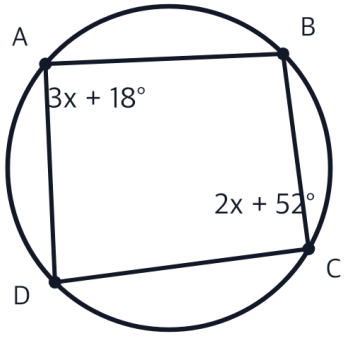
Show your working

Teacher answers & working

Circle Theorems · Algebra & Proof

GEO-475F-L3 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

SA1



ABCD is cyclic. Opposite angles are 3x + 18° and 2x + 52°. Find x.

Answer:

22

Working:

Opposite angles in a cyclic quadrilateral add to 180°.

$(3x + 18°) + (2x + 52°) = 180°$

$5x + 70 = 180$

$5x = 110$

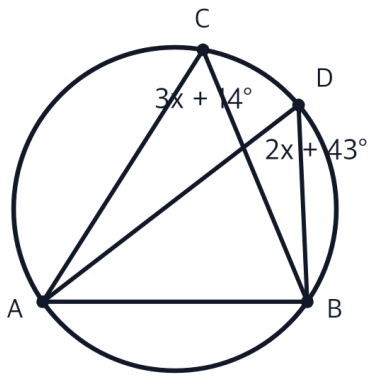
$x = 22$

Teacher answers & working

Circle Theorems · Algebra & Proof

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SB1



Angles ∠ACB and ∠ADB stand on chord AB. If ∠ACB = 3x + 14° and ∠ADB = 2x + 43°, find x.

Answer:

29

Working:

Angles in the same segment are equal.

$\angle ACB = \angle ADB$

$3x + 14° = 2x + 43°$

$3x + 14 = 2x + 43$

$x + 14 = 43$

$x = 29$

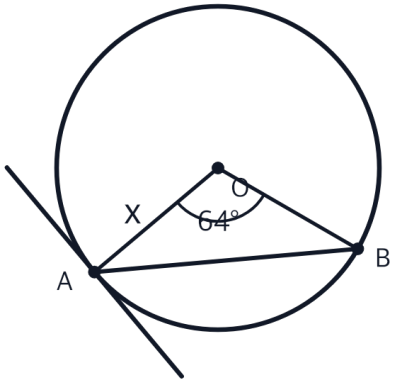
$x = 29$

Teacher answers & working

Circle Theorems · Algebra & Proof

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SC1



OA and OB are radii. The line at A is a tangent and ∠AOB = 64°. Find the angle between the tangent and chord AB.

Answer:

32°

Working:

OA and OB are radii, so triangle AOB is isosceles.

$\angle OAB = \angle OBA$.

The two base angles share 180° - 64°.

$\angle OAB = (180° - 64°) \div 2 = 58°$.

The tangent is perpendicular to radius OA.

$x + \angle OAB = 90°$

$x = 90° - 58°$

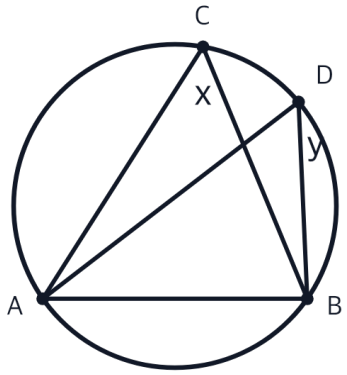
$x = 32°$

Teacher answers & working

Circle Theorems · Algebra & Proof

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SD1



Explain why ∠ACB = ∠ADB.

Answer:

They are equal: angles in the same segment.

Working:

Both ∠ACB and ∠ADB have their endpoints on A and B.

So both angles stand on the same chord AB.

Angles in the same segment are equal.

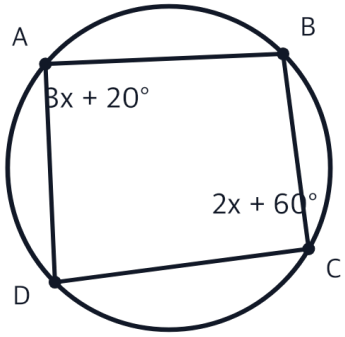
Therefore ∠ACB = ∠ADB.

Teacher answers & working

Circle Theorems · Algebra & Proof

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EA1



ABCD is cyclic. Opposite angles are 3x + 20° and 2x + 60°. Find x.

Answer:

20

Working:

Opposite angles in a cyclic quadrilateral add to 180°.

$(3x + 20°) + (2x + 60°) = 180°$

$5x + 80 = 180$

$5x = 100$

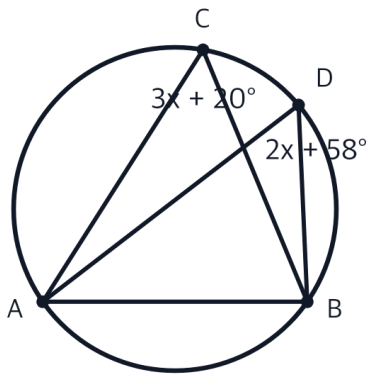
$x = 20$

Teacher answers & working

Circle Theorems · Algebra & Proof

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EB1



Angles ∠ACB and ∠ADB stand on chord AB. If ∠ACB = 3x + 20° and ∠ADB = 2x + 58°, find x.

Answer:

38

Working:

Angles in the same segment are equal.

$\angle ACB = \angle ADB$

$3x + 20° = 2x + 58°$

$3x + 20 = 2x + 58$

$x + 20 = 58$

$x = 38$

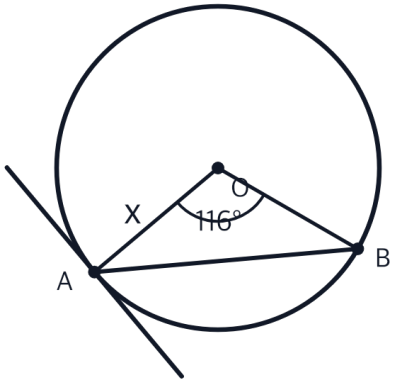
$x = 38$

Teacher answers & working

Circle Theorems · Algebra & Proof

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EC1



OA and OB are radii. The line at A is a tangent and ∠AOB = 116°. Find the angle between the tangent and chord AB.

Answer:

58°

Working:

OA and OB are radii, so triangle AOB is isosceles.

$\angle OAB = \angle OBA$.

The two base angles share 180° - 116°.

$\angle OAB = (180° - 116°) \div 2 = 32°$.

The tangent is perpendicular to radius OA.

$x + \angle OAB = 90°$

$x = 90° - 32°$

$x = 58°$